

Quiz 3-2 [25 min]

Check **ONLY T** if the given statement in the question is correct or **ONLY F** if not.

Correct Answer 4pts, No Answer -1 pts, Wrong Answer -2 pts. If the total points < 0, your score will be set to 0.

1. The following two propositions are logically equivalent: $p \rightarrow (\neg q \wedge r)$, $\neg p \vee \neg(r \rightarrow q)$. **(T)**
2. This proposition is a tautology: $((p \rightarrow \neg q) \wedge q) \rightarrow \neg p$. **(T)**
3. Suppose the variable x represents students and the variable y represents courses, and
 $A(y)$: y is an advanced course, $F(x)$: x is a freshman, $T(x, y)$: x is taking y , $P(x, y)$: x passed y .
By using the above predicates and any needed quantifiers, "No one is taking every advanced course." is written as
 $\neg \exists x \forall y (A(y) \rightarrow T(x, y))$. **(T)**
4. An argument of the following form is not valid: **(T)**

$$\begin{array}{l} p \rightarrow q \\ \neg p \\ \hline \therefore \neg q \end{array}$$

5. Suppose $g : A \rightarrow B$ and $f : B \rightarrow C$, where $f \circ g$ is 1-1 and f is 1-1. Then g be 1-1. **(T)**
6. Suppose $g : A \rightarrow B$ and $f : B \rightarrow C$, where $f \circ g$ is 1-1 and g is 1-1. Then f be 1-1. **(F)**
7. Suppose $g : A \rightarrow B$ and $f : B \rightarrow C$ where $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$, $C = \{2, 3, 6, 8\}$, and g and f are defined by $g = \{(a, 2), (b, 1), (c, 3), (d, 2)\}$ and $f = \{(1, 8), (2, 3), (3, 2)\}$. Then $f \circ g = \{(a, 3), (b, 8), (c, 2)\}$. **(F)**
8. The function $f : \mathbf{Z} \rightarrow \mathbf{Z}$ where $f(x) = x \bmod 10$ does not have its inverse. **(T)**
9. If A , B , and C are sets, then $A - (B \cap C) = (A - B) \cap (A - C)$. **(F)**
10. The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . **(T)**
11. $P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$. **(T)**
12. Two sets are equal if and only if they have the same cardinalities. **(F)**